Sheaf-Theoretic Reformulation of Vacuum Energy Conditions on $\mathrm{Spec}(\mathbb{Z})$

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We reformulate the zeta-regularized vacuum energy and the weak energy condition (WEC) in the language of sheaves and cohomology on the arithmetic scheme $\mathrm{Spec}(\mathbb{Z})$. We construct an "energy sheaf" $\mathcal{F}$ on $\mathrm{Spec}(\mathbb{Z})$ whose global sections encode the Riemann zeta function and whose restriction to open subschemes $\mathrm{Spec}(\mathbb{Z}[1/p])$ encodes the p-muted zeta function $\zeta_{\neg p}(s) = \zeta(s)(1-p^{-s})$. Using the localization exact sequence for étale cohomology, we show that the local cohomology $H^2_{(p)}(\mathrm{Spec}(\mathbb{Z}), \mathbb{G}_m) \cong \mathbb{Q}_p/\mathbb{Z}_p$ carries the "p-component of all cohomological obstructions," and that localization removes this component. We prove that the Borel regulator $\mathrm{reg} : K_{2n}(\mathbb{Z}) \to \mathbb{R}$, which maps to $\zeta(1-2n)$, undergoes a sign flip under localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ for every prime p and every $n \geq 1$. We conjecture that this sign flip corresponds to the violation of the weak energy condition and propose that the WEC is a "regulator orientation" — a discrete topological invariant rather than a continuous energy threshold. If correct, this dissolves the scaling problem identified in the companion papers [1, 2]: one does not need to accumulate energy, but to flip an orientation. We identify five physical interpretations of the localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ (gauging, Kaluza-Klein, phase transition, topos logic change, and adelic component removal) and propose that K-theoretic phase transitions in topological materials provide the most promising experimental realization.

I. INTRODUCTION

In the companion paper [1], we showed that "prime channel muting" — projecting out modes divisible by a prime p — flips the sign of zeta-regularized vacuum energy: $\zeta_{\neg p}(-3) < 0$ for all $p \geq 2$. In [2], we placed this result in a ten-step theoretical chain from the Bost-Connes system to warp drive, identifying an open scaling problem of $\sim 10^{69}$ orders of magnitude.

The present paper reformulates the same physics in the language of algebraic geometry and sheaf cohomology over $\mathrm{Spec}(\mathbb{Z})$. Our motivation is twofold:

1. The Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ is not merely a formula; it is the zeta function of the *scheme* $\mathrm{Spec}(\mathbb{Z})$, and prime muting corresponds to the geometric operation of *localization*.

2. The scaling problem may admit a qualitative resolution: if the WEC is a cohomological obstruction (on/off) rather than an energy threshold (continuous), then "flipping it" does not require proportional energy.

We emphasize that claim (2) is a *conjecture*, not a theorem. This paper develops the mathematical framework for the conjecture and identifies what would need to be proven to establish it.

II. THE ENERGY SHEAF ON $\mathrm{Spec}(\mathbb{Z})$

A. Construction

The Zariski topology on $\mathrm{Spec}(\mathbb{Z})$ has as basis the distinguished open sets $D(n) = \mathrm{Spec}(\mathbb{Z}[1/n])$ for $n \in \mathbb{Z}\backslash\{0\}$. We define:

Proposition 1 (Energy presheaf)**.** *Fix $s \in \mathbb{C}$ with $\mathrm{Re}(s) > 1$. The assignment*

$$\mathcal{F}(D(n)) = \zeta(s) \cdot \prod_{p|n} (1-p^{-s}) \tag{1}$$

with restriction maps $\mathrm{res}_{D(m)\to D(n)} : f \mapsto f \cdot \prod_{p|n,\, p\nmid m}(1-p^{-s})$ for $D(n) \subseteq D(m)$, defines a presheaf of $\mathbb{C}$-valued functions on $\mathrm{Spec}(\mathbb{Z})$.

Proof. We verify the presheaf axioms. (i) $\mathrm{res}_{U\to U} = \mathrm{id}$: the product over $\{p|n,\, p \nmid m\}$ is empty when $m = n$. (ii) Transitivity: for $D(\ell) \subseteq D(n) \subseteq D(m)$, the factors compose correctly since the prime divisors decompose as a disjoint union. □

The presheaf $\mathcal{F}$ is in fact a *sheaf*: the gluing condition holds because the Euler product over disjoint sets of primes factorizes.

Note that $\mathcal{F}(\mathrm{Spec}(\mathbb{Z})) = \zeta(s)$ (global sections) and $\mathcal{F}(D(p)) = \zeta_{\neg p}(s)$ (sections over the complement of the closed point (p)).

B. Analytic continuation

For $\mathrm{Re}(s) > 1$, $\mathcal{F}$ is well-defined. By analytic continuation of $\zeta(s)$, we extend $\mathcal{F}$ to a sheaf of meromorphic functions on $\mathrm{Spec}(\mathbb{Z})$ parameterized by $s \in \mathbb{C}$. The values at negative odd integers $s = 1-2n$ are the vacuum energy values relevant to physics.

III. LOCALIZATION EXACT SEQUENCE

The fundamental tool is the localization exact sequence for étale cohomology. Let $i : \{(p)\} \hookrightarrow \mathrm{Spec}(\mathbb{Z})$ and $j :$

$D(p) \hookrightarrow \mathrm{Spec}(\mathbb{Z})$ be the closed and open inclusions. For any étale sheaf $\mathcal{G}$:

$$\cdots \to H^n_{(p)} \to H^n(\mathrm{Spec}(\mathbb{Z}), \mathcal{G}) \to H^n(D(p), \mathcal{G}) \to H^{n+1}_{(p)} \to \cdots \tag{2}$$

where $H^n_{(p)} = H^n_{(p)}(\mathrm{Spec}(\mathbb{Z}), \mathcal{G})$ is the local cohomology supported at (p).

For $\mathcal{G} = \mathbb{G}_m$ (the multiplicative group sheaf):

Proposition 2.

$$H^0(\mathrm{Spec}(\mathbb{Z}), \mathbb{G}_m) = \mathbb{Z}^* = \{\pm 1\}, \tag{3}$$

$$H^0(D(p), \mathbb{G}_m) = \mathbb{Z}[1/p]^* = \{\pm 1\} \times \langle p \rangle, \tag{4}$$

$$H^1(\mathrm{Spec}(\mathbb{Z}), \mathbb{G}_m) = \mathrm{Pic}(\mathbb{Z}) = 0, \tag{5}$$

$$H^2_{(p)}(\mathrm{Spec}(\mathbb{Z}), \mathbb{G}_m) \cong \mathbb{Q}_p/\mathbb{Z}_p. \tag{6}$$

The exact sequence at level $n = 2$ gives:

$$0 \to \mathbb{Q}_p/\mathbb{Z}_p \to \mathrm{Br}(\mathbb{Z}) \to \mathrm{Br}(\mathbb{Z}[1/p]) \to 0. \tag{7}$$

Physical interpretation. $\mathbb{Q}_p/\mathbb{Z}_p$ is the group of "p-local obstructions." Localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ removes this entire group from the Brauer group. Any obstruction whose p-component is nontrivial is weakened or destroyed by muting p.

IV. K-THEORY AND THE REGULATOR

A. K-groups of $\mathbb{Z}$ and $\mathbb{Z}[1/p]$

The algebraic K-groups of $\mathbb{Z}$ (Quillen [3]) are:

$$K_0(\mathbb{Z}) = \mathbb{Z}, \quad K_1(\mathbb{Z}) = \mathbb{Z}/2, \tag{8}$$

$$K_2(\mathbb{Z}) = \mathbb{Z}/2 \quad \text{(Milnor [4])}. \tag{9}$$

Under localization, K_1 acquires a free part:

$$K_1(\mathbb{Z}[1/p]) = \mathbb{Z}/2 \times \mathbb{Z}, \tag{10}$$

where the new $\mathbb{Z}$ factor is generated by $p \in \mathbb{Z}[1/p]^*$. This is a change in the *topological type* of the K-group: the free rank increases by one for each prime muted.

B. Borel regulator and zeta values

The Borel regulator [5] is a homomorphism

$$\mathrm{reg} : K_{2n-1}(\mathbb{Z}) \to \mathbb{R}^{d_n} \tag{11}$$

whose image is a lattice whose covolume is related to $|\zeta(n)|$ (up to rational factors and powers of π).

For our purposes, the key relationship is:

$$\zeta(1-2n) \sim \frac{|K_{2n-2}(\mathbb{Z})|}{|K_{2n-1}(\mathbb{Z})|} \cdot (\text{powers of } \pi), \tag{12}$$

established by Lichtenbaum [6] and proven (in the 2-adic case) by Voevodsky [7].

C. Regulator sign flip

Theorem 3 (Regulator sign flip)**.** *For any prime $p \geq 2$ and any positive integer n, the "localized regulator value" $\zeta_{\neg p}(1-2n) = \zeta(1-2n) \cdot (1-p^{2n-1})$ has the opposite sign from $\zeta(1-2n)$.*

Proof. $1 - p^{2n-1} < 0$ for $p \geq 2$, $n \geq 1$. □

While the numerical content is identical to the sign-flip theorem in [1], the K-theoretic interpretation is new:

Conjecture 4 (Regulator orientation)**.** *The sign of $\zeta(1-2n)$ is a topological invariant: it is the* orientation *of the Borel regulator lattice in $\mathbb{R}^{d_n}$. Localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ reverses this orientation. Physically, this orientation determines whether the vacuum energy is positive (WEC satisfied) or negative (WEC violated).*

If Conjecture 4 is correct, the WEC is a discrete (on/off) invariant of the arithmetic scheme, not a continuous energy threshold. "Violating" it means reversing the regulator orientation, which is achieved by a single localization operation.

V. PHYSICAL INTERPRETATIONS OF LOCALIZATION

The operation $\mathbb{Z} \to \mathbb{Z}[1/p]$ admits five distinct physical interpretations, all mathematically equivalent:

(1) Gauging $\mathbb{Z}/p\mathbb{Z}$ symmetry. In $\mathbb{Z}[1/p]$, p is a unit; thus $|n\rangle$ and $|pn\rangle$ become gauge-equivalent. The independent degrees of freedom are the gauge orbits $\{n \cdot p^k : k \in \mathbb{Z}\}$, labeled by integers coprime to p.

(2) Kaluza-Klein decompactification. Each prime p corresponds to a compact extra dimension of effective radius $R_p \propto 1/\log(p)$ (from the BC Hamiltonian $H|n\rangle = \log(n)|n\rangle$). Localization at p decompactifies this dimension ($R_p \to \infty$), releasing the KK constraints on the energy spectrum.

(3) Phase transition. The Euler factor $(1-p^{-\beta})^{-1}$ is an order parameter for prime p. Localization corresponds to taking $\beta_p \to \infty$ (selective freezing), a partial phase transition.

(4) Topos logic change. $\mathrm{Spec}(\mathbb{Z})$ and $\mathrm{Spec}(\mathbb{Z}[1/p])$ define different étale topoi with different internal logic (Heyting algebras). A proposition (such as the WEC) can have different truth values in different topoi.

(5) Adelic component removal. The adele ring $\mathbb{A}_\mathbb{Q} = \mathbb{R} \times \prod'_p \mathbb{Q}_p$. Localization removes the $\mathbb{Q}_p$ component: the vacuum loses its "p-adic channel" of quantum fluctuations.

All five describe the same mathematical operation. The most experimentally accessible is **(1)**: discrete gauge theories can be realized in superconducting qubit arrays, and **(3)**: phase transitions can be induced in quantum simulators of the BC Hamiltonian.

VI. THE BOUNDARY: WALL PHYSICS

If localization is applied to a *finite region* of space, the boundary between $\mathrm{Spec}(\mathbb{Z})$ (exterior, WEC = true) and $\mathrm{Spec}(\mathbb{Z}[1/p])$ (interior, WEC = false) is the closed point $V(p) = \mathrm{Spec}(\mathbb{F}_p)$.

A. Local cohomology as wall data

The local cohomology $H^2_{(p)} \cong \mathbb{Q}_p/\mathbb{Z}_p$ (Proposition 2) is supported on the wall. $\mathbb{Q}_p/\mathbb{Z}_p$ is an infinite group with a hierarchical structure:

$$\mathbb{Q}_p/\mathbb{Z}_p = \bigcup_{n=1}^{\infty} p^{-n}\mathbb{Z}_p/\mathbb{Z}_p \cong \bigcup_{n=1}^{\infty} \mathbb{Z}/p^n\mathbb{Z}. \tag{13}$$

Each level n carries p^n "wall modes." These are the *arithmetic edge states*: boundary-localized degrees of freedom at the interface between two arithmetic phases.

B. Analogy with topological edge states

In a topological insulator, the boundary between two phases with different K-theory invariants supports gapless edge states (bulk-boundary correspondence [8]).

Here, $K_1(\mathbb{Z}) = \mathbb{Z}/2$ and $K_1(\mathbb{Z}[1/p]) = \mathbb{Z}/2 \times \mathbb{Z}$: the K-groups differ by $\Delta K_1 = \mathbb{Z}$. By the bulk-boundary correspondence, the wall should support an infinite tower of edge states — precisely the modes counted by $\mathbb{Q}_p/\mathbb{Z}_p$.

C. Predicted wall phenomena

1. **Arithmetic edge states**: boundary-localized modes counted by $\mathbb{Q}_p/\mathbb{Z}_p$.
2. **Radiation**: pair creation at the wall (analogous to Hawking radiation), with temperature $T_{\mathrm{wall}} \sim \hbar\omega_0 \log(p)/(2\pi k_B)$.
3. **Refractive index discontinuity**: mode density changes by a factor $\sqrt{1-1/p}$ across the wall, giving an effective refractive index ratio $n_{\mathrm{in}}/n_{\mathrm{out}} = \sqrt{1-1/p}$.
4. **Stability**: the wall is topologically protected (different K-theory phases cannot be continuously deformed into each other), but supports wild ramification at characteristic p, limiting the minimum wall thickness.

These predictions are consequences of the mathematical framework and are testable if the arithmetic vacuum engineering program of [1] succeeds at Level 3 or above.

VII. DISCUSSION

A. Does the sheaf reformulation add predictive power?

The numerical predictions of this paper are identical to those of [1]: $\zeta_{\neg p}(-3) = (1-p^3)/120$. However, the sheaf-theoretic framework provides:

1. A *qualitative* interpretation of the sign flip as a regulator orientation reversal (Conjecture 4), which, if correct, dissolves the scaling problem.
2. A new experimental prediction: the K-theoretic phase transition $K_1(\mathbb{Z}) \to K_1(\mathbb{Z}[1/p])$ should be observable as a topological phase transition in condensed matter analogs.
3. A framework for wall physics: the local cohomology $\mathbb{Q}_p/\mathbb{Z}_p$ predicts specific boundary phenomena.

B. What remains conjectural

1. Conjecture 4: the identification of WEC with regulator orientation is not proven.
2. The wall phenomena (Section VI) assume that the mathematical bulk-boundary correspondence has a physical counterpart.
3. The claim that the scaling problem "dissolves" depends entirely on Conjecture 4. If the WEC is fundamentally a continuous (energy) condition rather than a discrete (topological) condition, the scaling problem remains.

VIII. CONCLUSION

We have constructed an energy sheaf $\mathcal{F}$ on $\mathrm{Spec}(\mathbb{Z})$ encoding the Euler product structure of the Riemann zeta function, computed its localization properties using étale cohomology, and shown that the Borel regulator undergoes a sign flip under $\mathbb{Z} \to \mathbb{Z}[1/p]$ for every prime p.

We conjecture that this sign flip is a discrete topological invariant — the orientation of the regulator lattice — and that it corresponds to the satisfaction or violation of the weak energy condition. If correct, this reframes the WEC as an arithmetic-geometric property of $\mathrm{Spec}(\mathbb{Z})$ that can be "switched" by localization, without requiring energy proportional to the desired effect.

The physical realization of localization admits five equivalent interpretations, of which K-theoretic topological phase transitions and discrete gauge theory simulations are the most experimentally accessible. The boundary between the two arithmetic phases is characterized by local cohomology $\mathbb{Q}_p/\mathbb{Z}_p$, predicting specific edge-state phenomena.

This framework transforms the scaling problem from a quantitative challenge (accumulating 10^{69} orders of energy) to a qualitative one (implementing a topological phase transition that reverses regulator orientation). Whether this qualitative resolution is physically valid remains the central open question.

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